R7.1: complete-cost ordering of kernel-finding routes

Round 7, 2026-10-31. Code `examples/r7_1_complete_cost.rs` and the library change in `src/cryptanalysis/isogeny_eval/` (native Rust). Raw records: `ladder_grid_*.json`, `ladder_extra_a.json`, `project_*.json`, `psi_*.json`, `modpoly_*.json`, `selftest.json`; fits and tables: `analysis.json`, `analysis.txt`; full write-up: `RESULTS.md`. Toy, synthetic instances only ($j = -3375$, $D = -7$, 61-bit p, one instance per ℓ); counted unit, not time; no sub-rho claim and no claim about a deployed curve.

Question, hypothesis, falsifier (from the task)

R5.3 and R6.4 priced rows as $M + W + 91I$ only. **H**: the ordering of the routes (ladder/NTT, ψ_ℓ + factoring, $\Phi_\ell(j, Y)$) and the crossover with rho at $r = 2^{128}$ survive when additions and Φ_ℓ construction are counted, with an addition priced at $a \in \{0, 0.1, 0.3, 1\}$ M. **Falsifier**: the ordering of the cheapest two routes changes at some a for some measured ℓ, or the exponent CI moves its lower endpoint below 1.5 or its upper above 2.3.

Verdict

The ordering part of the falsifier is triggered; H is not confirmed as stated. The exponent part is not triggered.

- Ordering. Cold accounting (Φ_ℓ construction charged), ascending edges: the second place between ψ and Φ swaps with a at $\ell = 89, 109, 131$ (3 of 22; the two come within 1.6%, 0.3%, 0.4% of each other over the weights, one instance per ℓ: a near tie). The ladder is cheapest at every ℓ and a in the cold accounting. Amortised (Φ_ℓ given), both directions, and cold descending: no change of the cheapest two or of the full order with a at any measured ℓ (order: amortised Φ < ladder < ψ; cold at $a = 0$: ladder < Φ < ψ).
- Exponent. Projected generic ladder ($\ell \geq 2^{13}$, `PolyMul::Auto`): 1.919 at every a; 95% OLS interval at worst [1.717, 2.121], bootstrap at worst [1.781, 2.126]; inside [1.5, 2.3]. At $a = 0$ this is R6.4′s 1.919 [1.724, 2.115]: still insufficient to show an exponent below 2.
- Crossover (supplementary, extrapolated; $r = 2^{128}$, $g = 6$): ladder $\log_2 \ell^*$ shifts by -0.24 to -0.39 bit from $a = 0$ to $a = 1$; ψ -0.37; Φ cold -0.01; Φ amortised -1.48 (fit over $\ell \leq 103$ extrapolated to 2^{30}). The rank order of the routes by ℓ^* does not change with a.

Method

Library: `PrimeField::counted_with_adds` tallies additions in `AddTally` (opt-in; M, W, I, `OpTally` and all frozen outputs unchanged). One addition = one modular add/sub/neg: every `add`/`sub`/`neg` call; schoolbook accumulations $l_a l_b - (l_a + l_b - 1)$; per NTT product $3TN \log_2 N + 3n$ + data-dependent conditional subtractions modulo NTT primes, and $2n$ `u128` accumulations in F_p. Cost $= M + W + 91I + a(A + WA)$ with $W = 1M$ as before. Route (c) now includes Φ_ℓ construction (counted port of the library's q-series builder; equal to the library's table for all 23 $\ell \leq 103$; fast and per-operation ports agree in table and tally) and the Elkies kernel polynomial (counted port; equal to the library's for all 28 ℓ, both directions); reported cold (construction charged per edge) and amortised (Φ_ℓ given). Sutherland's CRT evaluation is **not implemented**. Arms (a) ladder over F_{p^k}, (a′) per-step projection to $\ell = 2^{16}$, (b) ψ_ℓ + Cantor-Zassenhaus / gcd, on R6.4′s instances. Frozen counts are unchanged: the unmodified round-6 example rebuilt on the new library reproduces its records (9/9, 2/2, 5/5 rows), and this example's $a = 0$ columns reproduce R6.4′s counts (ladder 20/20, ψ 9/9, Φ 23/23, projection 119/119).

Exponents per route and weight (measured; (a') projected)

$\log_2 \text{cost} = \log_2 c + e \log_2 \ell$; intervals 95%: OLS t-interval and pairs bootstrap (2000 resamples).

series	$a = 0$	$a = 0.1$	$a = 0.3$	$a = 1$
(a') projected ladder, Auto, $\ell \geq 2^{13}$	1.919 OLS [1.724, 2.115] boot [1.795, 2.113]	1.919 OLS [1.722, 2.116] boot [1.781, 2.113]	1.919 OLS [1.720, 2.118] boot [1.790, 2.126]	1.919 OLS [1.717, 2.121] boot [1.783, 2.125]
(a') projected ladder, Auto, $\ell \geq 2^{10}$	2.088 OLS [2.020, 2.155] boot [2.023, 2.158]	2.092 OLS [2.024, 2.160] boot [2.029, 2.163]	2.097 OLS [2.028, 2.167] boot [2.029, 2.169]	2.105 OLS [2.034, 2.177] boot [2.035, 2.179]
(a) ladder measured, desc, $\ell \geq 256$	2.554 OLS [2.352, 2.756] boot [2.376, 2.713]	2.566 OLS [2.380, 2.753] boot [2.399, 2.710]	2.584 OLS [2.420, 2.748] boot [2.434, 2.715]	2.612 OLS [2.482, 2.742] boot [2.498, 2.720]
(b) ψ + CZ, desc, 37..131	3.565 OLS [3.302, 3.828] boot [3.316, 3.792]	3.588 OLS [3.331, 3.845] boot [3.346, 3.819]	3.618 OLS [3.370, 3.867] boot [3.384, 3.833]	3.665 OLS [3.428, 3.903] boot [3.454, 3.856]
(b) ψ + gcd, asc, 37..131	3.439 OLS [3.087, 3.791] boot [3.187, 3.724]	3.456 OLS [3.106, 3.806] boot [3.222, 3.738]	3.478 OLS [3.130, 3.826] boot [3.228, 3.743]	3.512 OLS [3.166, 3.859] boot [3.270, 3.764]
(c) Φ amortised, desc, 11..103	1.860 OLS [1.785, 1.936] boot [1.779, 1.932]	1.882 OLS [1.808, 1.956] boot [1.804, 1.959]	1.914 OLS [1.842, 1.987] boot [1.838, 1.988]	1.976 OLS [1.906, 2.045] boot [1.900, 2.043]
(c) Φ cold, desc, 11..103	4.638 OLS [4.581, 4.696] boot [4.574, 4.751]	4.639 OLS [4.582, 4.696] boot [4.577, 4.750]	4.640 OLS [4.583, 4.697] boot [4.572, 4.753]	4.642 OLS [4.586, 4.698] boot [4.576, 4.745]
(c) Φ construction alone, 11..103	4.685 OLS [4.641, 4.729] boot [4.632, 4.768]	4.685 OLS [4.641, 4.729] boot [4.635, 4.769]	4.685 OLS [4.641, 4.729] boot [4.636, 4.771]	4.685 OLS [4.641, 4.729] boot [4.634, 4.768]

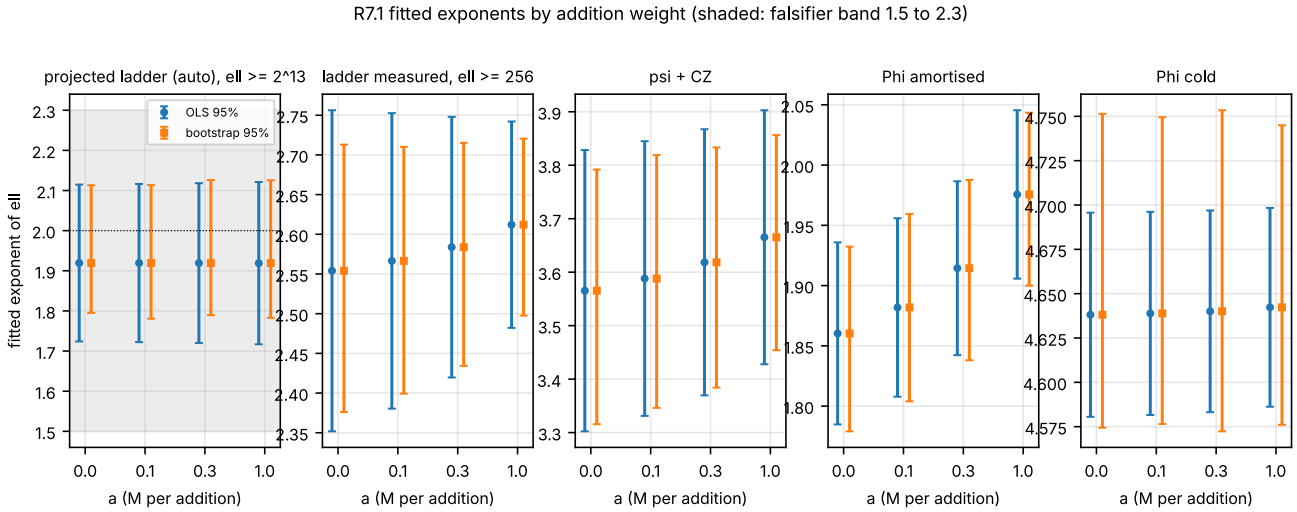


Figure 1: Fitted exponents by addition weight with OLS and bootstrap 95% intervals (data: `analysis.json`, `fits`; n as in the table).

Ordering of routes

L ladder, P ψ + factoring, F Φ amortised, C Φ cold. Counts over all measured ℓ with the three routes of the accounting measured (`analysis.json`, `ordering_summary`).

direction	accounting	ℓ tested	cheapest-two changes with a	smallest adjacent cost ratio
descending	amortised	25	0	4.670
descending	cold	25	0	1.605
ascending	amortised	22	0	11.635
ascending	cold	22	3 (at ℓ = 89, 109, 131)	1.003

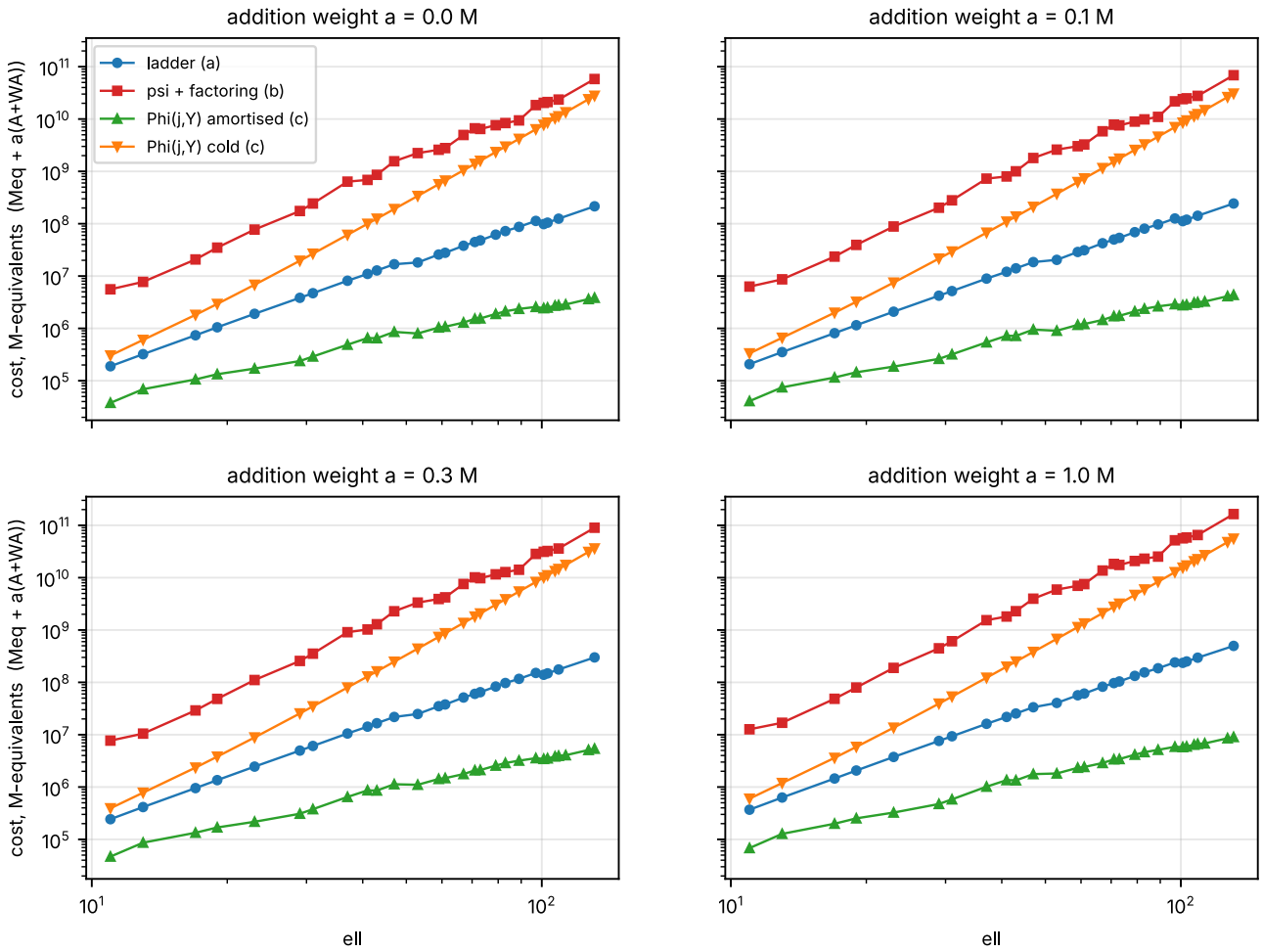


Figure 2: Measured cost of the four route accountings, descending edges, by addition weight (`analysis.json`, `ordering`; one instance per ℓ, counted unit).

Selected rows of the ordering table (cost in M-equivalents, $M + W + 91I + a(A + WA)$; full table in `RESULTS.md` and `analysis.json`):

ℓ	dir	a	L	P	F	C	order amortised	order cold
37	desc	0.0	8.13e+06	6.37e+08	4.88e+05	6.13e+07	F < L < P	L < C < P
37	desc	0.1	8.93e+06	7.27e+08	5.40e+05	6.74e+07	F < L < P	L < C < P
37	desc	0.3	1.05e+07	9.08e+08	6.46e+05	7.97e+07	F < L < P	L < C < P
37	desc	1.0	1.61e+07	1.54e+09	1.02e+06	1.23e+08	F < L < P	L < C < P
37	asc	0.0	8.14e+06	3.39e+08	2.03e+05	6.10e+07	F < L < P	L < C < P
37	asc	0.1	8.94e+06	3.89e+08	2.24e+05	6.71e+07	F < L < P	L < C < P
37	asc	0.3	1.05e+07	4.88e+08	2.67e+05	7.93e+07	F < L < P	L < C < P
37	asc	1.0	1.62e+07	8.37e+08	4.17e+05	1.22e+08	F < L < P	L < C < P
59	desc	0.0	2.58e+07	2.59e+09	1.04e+06	5.68e+08	F < L < P	L < C < P
59	desc	0.1	2.89e+07	3.03e+09	1.17e+06	6.24e+08	F < L < P	L < C < P
59	desc	0.3	3.50e+07	3.91e+09	1.43e+06	7.38e+08	F < L < P	L < C < P
59	desc	1.0	5.67e+07	7.00e+09	2.33e+06	1.14e+09	F < L < P	L < C < P
59	asc	0.0	2.58e+07	1.24e+09	3.76e+05	5.67e+08	F < L < P	L < C < P
59	asc	0.1	2.89e+07	1.46e+09	4.23e+05	6.24e+08	F < L < P	L < C < P
59	asc	0.3	3.51e+07	1.90e+09	5.16e+05	7.37e+08	F < L < P	L < C < P
59	asc	1.0	5.68e+07	3.42e+09	8.41e+05	1.13e+09	F < L < P	L < C < P
79	desc	0.0	6.18e+07	7.62e+09	1.89e+06	2.34e+09	F < L < P	L < C < P
79	desc	0.1	6.89e+07	8.94e+09	2.11e+06	2.57e+09	F < L < P	L < C < P
79	desc	0.3	8.31e+07	1.16e+10	2.56e+06	3.04e+09	F < L < P	L < C < P
79	desc	1.0	1.33e+08	2.08e+10	4.14e+06	4.67e+09	F < L < P	L < C < P
79	asc	0.0	6.19e+07	3.53e+09	6.60e+05	2.34e+09	F < L < P	L < C < P
79	asc	0.1	6.90e+07	4.16e+09	7.37e+05	2.57e+09	F < L < P	L < C < P
79	asc	0.3	8.32e+07	5.42e+09	8.93e+05	3.04e+09	F < L < P	L < C < P
79	asc	1.0	1.33e+08	9.80e+09	1.44e+06	4.67e+09	F < L < P	L < C < P
97	desc	0.0	1.13e+08	1.85e+10	2.57e+06	6.37e+09	F < L < P	L < C < P
97	desc	0.1	1.26e+08	2.18e+10	2.90e+06	7.00e+09	F < L < P	L < C < P
97	desc	0.3	1.51e+08	2.84e+10	3.56e+06	8.28e+09	F < L < P	L < C < P
97	desc	1.0	2.40e+08	5.18e+10	5.87e+06	1.27e+10	F < L < P	L < C < P

97	asc	0.0	1.13e+08	9.11e+09	8.62e+05	6.36e+09	F < L < P	L < C < P
97	asc	0.1	1.26e+08	1.08e+10	9.73e+05	7.00e+09	F < L < P	L < C < P
97	asc	0.3	1.52e+08	1.41e+10	1.20e+06	8.27e+09	F < L < P	L < C < P
97	asc	1.0	2.41e+08	2.56e+10	1.98e+06	1.27e+10	F < L < P	L < C < P
103	desc	0.0	1.05e+08	2.11e+10	2.49e+06	8.54e+09	F < L < P	L < C < P
103	desc	0.1	1.19e+08	2.49e+10	2.83e+06	9.40e+09	F < L < P	L < C < P
103	desc	0.3	1.49e+08	3.23e+10	3.52e+06	1.11e+10	F < L < P	L < C < P
103	desc	1.0	2.51e+08	5.85e+10	5.94e+06	1.71e+10	F < L < P	L < C < P
103	asc	0.0	1.05e+08	9.78e+09	8.52e+05	8.54e+09	F < L < P	L < C < P
103	asc	0.1	1.20e+08	1.15e+10	9.71e+05	9.39e+09	F < L < P	L < C < P
103	asc	0.3	1.49e+08	1.51e+10	1.21e+06	1.11e+10	F < L < P	L < C < P
103	asc	1.0	2.51e+08	2.74e+10	2.05e+06	1.71e+10	F < L < P	L < C < P
109	desc	0.0	1.25e+08	2.35e+10	2.77e+06	1.13e+10	F < L < P	L < C < P
109	desc	0.1	1.42e+08	2.77e+10	3.16e+06	1.24e+10	F < L < P	L < C < P
109	desc	0.3	1.76e+08	3.61e+10	3.94e+06	1.47e+10	F < L < P	L < C < P
109	desc	1.0	2.96e+08	6.53e+10	6.66e+06	2.25e+10	F < L < P	L < C < P
109	asc	0.0	1.25e+08	1.05e+10	9.61e+05	1.13e+10	F < L < P	L < P < C
109	asc	0.1	1.42e+08	1.24e+10	1.09e+06	1.24e+10	F < L < P	L < P < C
109	asc	0.3	1.76e+08	1.61e+10	1.36e+06	1.47e+10	F < L < P	L < C < P
109	asc	1.0	2.96e+08	2.93e+10	2.29e+06	2.25e+10	F < L < P	L < C < P
131	desc	0.0	2.15e+08	5.82e+10	3.84e+06	2.78e+10	F < L < P	L < C < P
131	desc	0.1	2.43e+08	6.88e+10	4.36e+06	3.06e+10	F < L < P	L < C < P
131	desc	0.3	3.00e+08	8.99e+10	5.39e+06	3.62e+10	F < L < P	L < C < P
131	desc	1.0	4.97e+08	1.64e+11	9.02e+06	5.57e+10	F < L < P	L < C < P
131	asc	0.0	2.15e+08	2.60e+10	1.38e+06	2.78e+10	F < L < P	L < P < C
131	asc	0.1	2.44e+08	3.07e+10	1.56e+06	3.06e+10	F < L < P	L < C < P
131	asc	0.3	3.00e+08	4.02e+10	1.92e+06	3.62e+10	F < L < P	L < C < P
131	asc	1.0	4.98e+08	7.35e+10	3.19e+06	5.56e+10	F < L < P	L < C < P

Route (c): what construction costs

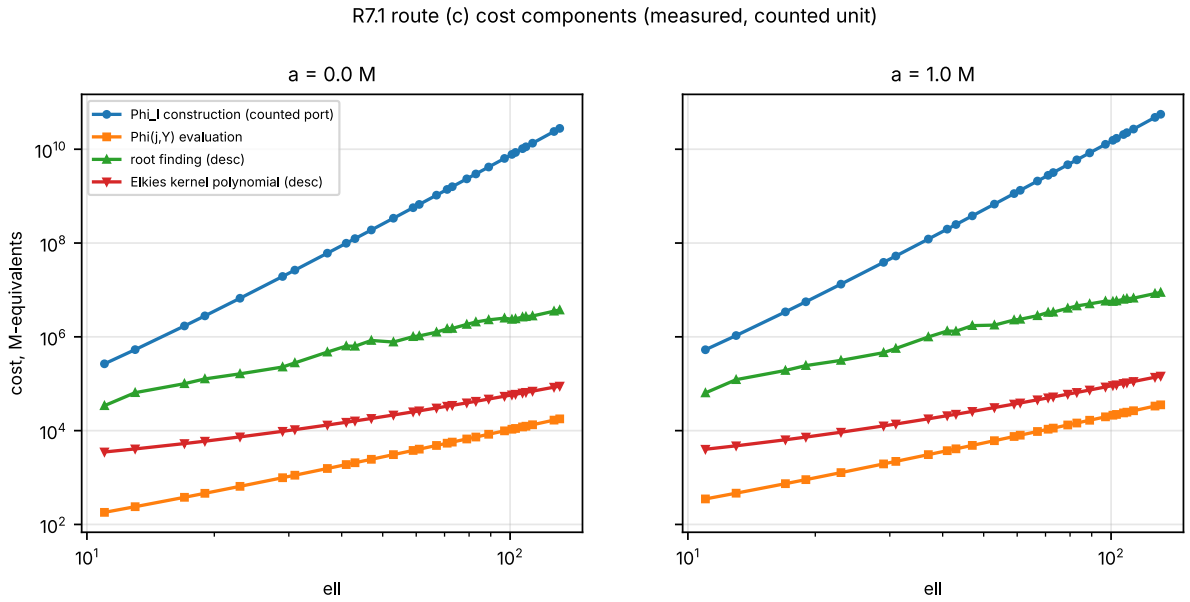


Figure 3: Cost components of route (c) at $a = 0$ and $a = 1$ (`modpoly_*.json`): construction $\approx \ell^{4.69}$, evaluation, root finding, Elkies.

Construction / ladder find at $a = 0$: 7.5 ($\ell = 37$), 28 (67), 81 (103), 129 (131). Additions per M-equivalent: construction 1.00, ladder 0.99 ($\ell = 37$) to 1.61 (863), ψ 1.4 to 1.8, projected NTT ladder 1.8.

Crossover with rho (extrapolated), $r = 2^{128}$, $g = 6$, $\log_2 \ell^*$

rho cost $\sqrt{\pi \frac{r}{4}}(g + ga)$ (g multiplications and g additions per group operation, stated assumption); find costs scaled by $\log_2 \frac{r}{61}$; $\ell_{\max} = 2^{65}$.

model	$a = 0$	$a = 0.1$	$a = 0.3$	$a = 1$	shift
ladder (a), measured desc fit l≥256	21.74	21.66	21.54	21.35	−0.39
ladder (a), measured asc fit l≥256	21.74	21.66	21.54	21.35	−0.39
ladder (a'), projected auto fit l≥2^10	24.23	24.16	24.07	23.92	−0.31
ladder (a'), projected auto fit l≥2^13	25.07	25.02	24.94	24.81	−0.25
ladder (a'), projected auto l≥2^13, exponent at OLS CI low end	26.10	26.05	25.98	25.85	−0.24
ladder (a'), projected best of karatsuba/auto fit l≥2^13	25.07	25.02	24.94	24.81	−0.25
ladder (a'), model l^2 log l calibrated at top	24.55	24.51	24.44	24.32	−0.23
psi + CZ desc (b), measured fit	15.38	15.30	15.19	15.01	−0.37
psi + gcd asc (b), measured fit	16.02	15.95	15.84	15.69	−0.34

Phi amortised desc (c), measured fit 11..103	30.26	29.97	29.54	28.77	−1.48
Phi cold desc (c), measured fit 11..103	13.69	13.69	13.69	13.68	−0.01

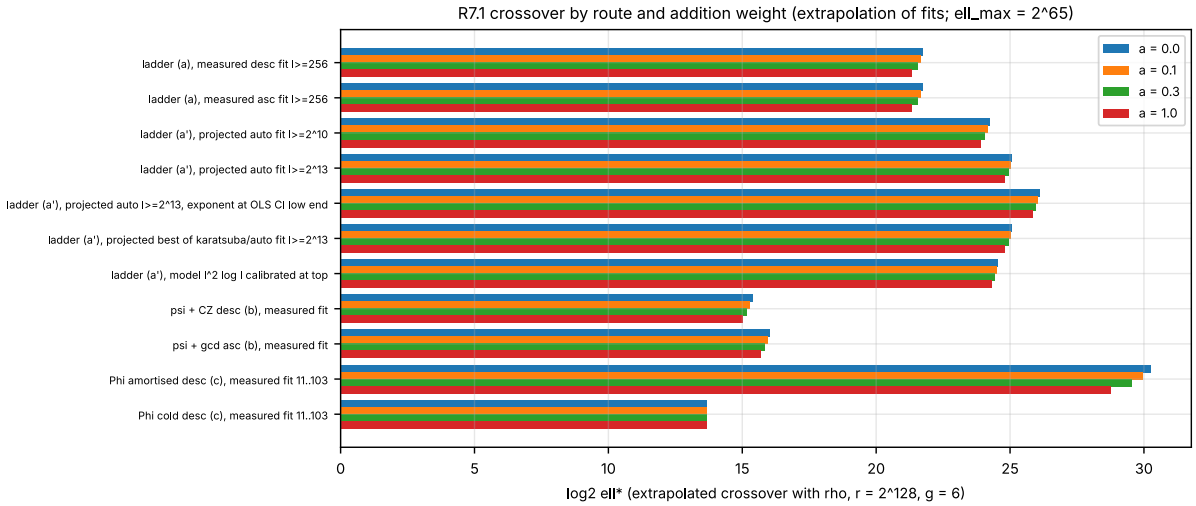


Figure 4: Extrapolated crossover with rho by route and weight (`analysis.json`, `crossover`).

Limits (what is still not counted)

Sutherland's CRT evaluation (cold route (c) is the library builder, exponent 4.69); setup of F_{p^k} and group orders for the ladder, kernel certification; additions inside inversions ($I = 91M$ fixed), reductions, memory and storage of the Φ_ℓ table, any wall time; `PolyMul::Auto` chooses by $M + W$ (whole-ladder best-of Karatsuba/Auto checked, equal to Auto for $\ell \geq 2^{13}$); a is swept, not measured on hardware; the rho step's g additions are an assumption; one instance per ℓ (the near ties are within that noise); crossovers and the projection beyond $\ell = 863$ are extrapolations; the full-crate library test was not run in this session (memory limit), the `isogeny_eval` tests (18) pass in a minimal crate that includes the same sources.